EFU FISHERY & AQUACULTURE LICENCE v1.0

Experimental EFU-based Sustainability & Sovereignty Framework

0. Purpose and Scope

This document establishes an experimental, EFU-based licensing and measurement framework for fishery and aquaculture activities.

It is **not** a regulation, mandatory industry standard, or legal requirement, but an open research protocol that enables marine and freshwater resource use to be measured and compared using human-scale EFU metrics (energy, water, materials, entropy).

I. Core Concepts

1. **EFU (Human Flux Unit)**
   Human-centric biophysical unit relating fishery and aquaculture systems' resource flows to an average adult's daily total material flux (~20 kg/day).
2. **EFU-Fish (EFU-F)**
   EFU framework application for fisheries/aquaculture, comprising:
   - **EFU-H_Fish**: energy use (fuel, electricity) per fish unit
   - **EFU-W_Fish**: water loading (nutrients, oxygen, water exchange)
   - **EFU-M_Fish**: material use (feed, nets, plastics)
   - **EFU-S_Fish**: entropy/waste heat, ecosystem disruption (experimental)
   - **EFU-C** (optional): carbon footprint per fish unit.
3. **Sovereignty Gap – Fishery Context**
   Difference between fishery/aquaculture EFU load and regional biocapacity-based EFU allocation (SS_Fish = TEFU_Fish – Allocated_EFU_Fish), consistent with general EFU-Sovereignty model.

II. EFU Fishery Licence – Research Status

1. **What the EFU-Fish Licence represents**
   The "EFU Fishery & Aquaculture Licence v1.0" documents that a fleet, farm, or region has undergone EFU audit with results documented in standardized, public, EFU-consistent format.
2. **What the Licence does NOT represent**
   - Not a legal fishing permit or quota
   - Does not replace fishery/aquaculture regulation, Natura 2000, MSC, or other certification systems
   - Does not guarantee profitability, market access, or "green" status

3. **What the Licence DOES represent**
    - Commitment to EFU-based measurement system
    - Documented, auditable key metrics (EFU-H_Fish, EFU-W_Fish, TEFU_Fish, SS_Fish, MROI_Fish)
    - Citable in EFU research context (case study, benchmark)

III. Metrics and Thresholds (Fishery-specific – experimental)

1. **EFU-H_Fish (energy per kg fish)**
    - EFU-H_Fish = (fuel_MWh + electricity_MWh) / (12.88 MWh/person/year) / fish_kg
    - Interpretation: human-year energy footprint per 1 kg fish
2. **EFU-W_Fish (water loading per kg fish)**
    - EFU-W_Fish = (nutrient + water exchange equivalent m^3) / 18.25 m^3/person/year / fish_kg
3. **TEFU_Fish – Total Fishery EFU Cost**
    - TEFU_Fish = w_H * EFU-H_Fish + w_W * EFU-W_Fish + w_M * EFU-M_Fish + w_S * EFU-S_Fish (+ w_C * EFU-C_Fish if relevant)
    - Weights (w_i) regionally calibratable (higher w_W in water-scarce regions)
4. **MROI_Fish – Metabolic Return on Investment (fishery)**
    - MROI_Fish = THI_Fish / TEFU_Fish
    - THI_Fish = composite of local food security, jobs, value added, ecosystem health
    - MROI bands for fisheries:
        - MROI_Fish > 1e-3 → high community benefit per EFU cost
        - 1e-4 – 1e-3 → acceptable, calibratable
        - <1e-5 → metabolically weak / overloading operation

IV. Licence Categories (EFU-Fish Labels – experimental)

1. **EFU-FISH SOVEREIGN**
    - Conditions:
        - SS_Fish ≤ 0 (total fishery/aquaculture TEFU_Fish within RCC_Fish allocation), **AND**
        - MROI_Fish in "high" or "acceptable" range (≥1e-4)
    - Meaning: metabolically sovereign, operating within regional biocapacity
2. **EFU-FISH BALANCED**
    - SS_Fish small positive range (0 < SS_Fish ≤ T_bal_Fish), MROI_Fish at least moderate
    - Meaning: balanced but requires continuous monitoring
3. **EFU-FISH DEFICIT**
    - SS_Fish significant positive range (T_bal_Fish < SS_Fish ≤ T_def_Fish), or MROI_Fish low
    - Meaning: system generates metabolic deficit, risky

4. **EFU-FISH CRITICAL**
    - SS_Fish > T_def_Fish, AND MROI_Fish very low (<1e-5)
    - Meaning: structurally imbalanced fishery system; critical case for EFU research

Note: Exact thresholds (T_bal_Fish, T_def_Fish) calibrated regionally/ecosystem-specifically in EFU-Fish v1.x supplements.

V. Branding & Integrity Rules (Fishery Extension)

1. **EFU-FISH Label usage**
    - "EFU-FISH SOVEREIGN / BALANCED / DEFICIT / CRITICAL" designation only after EFU-compatible audit (EFU-Fish protocol), per 104.3 EFU USE GUIDELINES spirit
2. **Prohibited misuse**
    - Prohibited to market fishery products/fleets as EFU-FISH labelled without documented EFU audit
    - Prohibited to use EFU-FISH labels to conceal overfishing or illegal activity; EFU framework does not legitimize violations

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VI. Integration & Feedback

- **Software**: EFU-Fish module extends EFU-Engine Python/Excel tools, same Apache 2.0 logic
- **Research status**: all EFU-Fish licences experimental; strongly recommended to share results on Zenodo/open data platforms (anonymized) for methodology calibration

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